

Slides: Restricted access to CS 422 Fall 2026; not to be shared online or in public forums

Slides adapted from Prof. Stefan Schmid's course at TU Berlin, and the Top-Down Approach textbook by Kurose-Ross

Overview & Circuit/Packet Switching

CS 422: Computer Networks

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What is the Internet, Anyway?



https://www.youtube.com/watch?v=UIJku_CSyNg&t=2s

What is the Internet, Anyway?

- RFC 1935: "What is the Internet, Anyway?"
- RFC: Request for Comments
- IETF: Internet Engineering Task Force
 - Protocol standardization

History of the Internet: Arpanet

- Started with UCLA, UCSB, SRI, and UTAH
- A network of computers at different universities

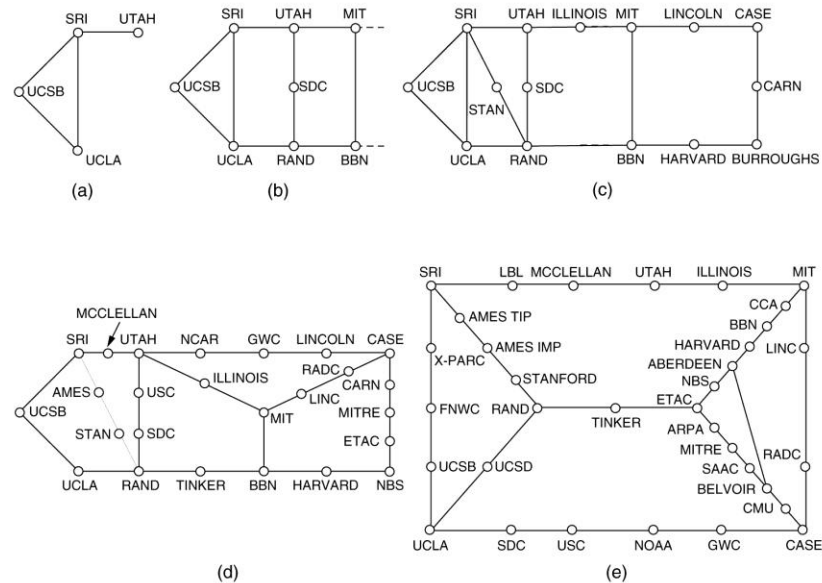


Image: computing.dcu.ie/~humphrys/Notes/Networks/tanenbaum/1-27.jpg

The Internet - General Architecture

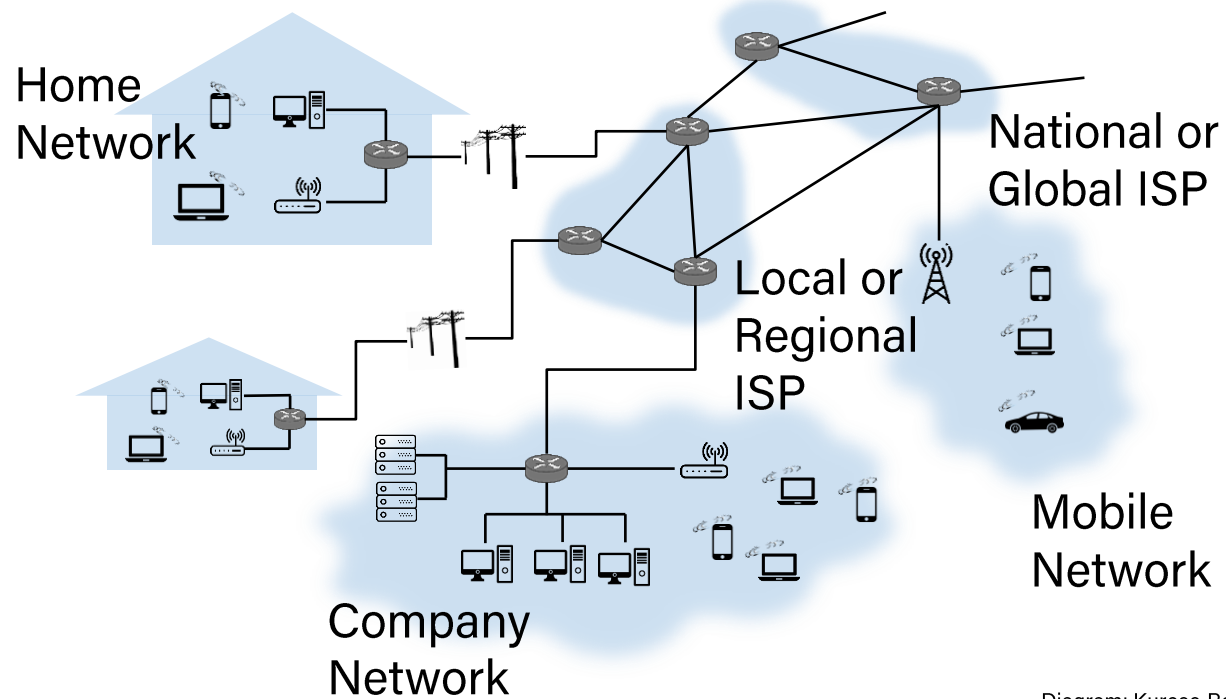


Diagram: Kurose-Ross

The Internet – General Architecture

- Switch: Forward traffic within a local network 
- Router: Route traffic through the network 
- Mobile tower: Wireless core access point 
- Modem: Provides home/wifi network 
- Server: Hosting services on the internet 

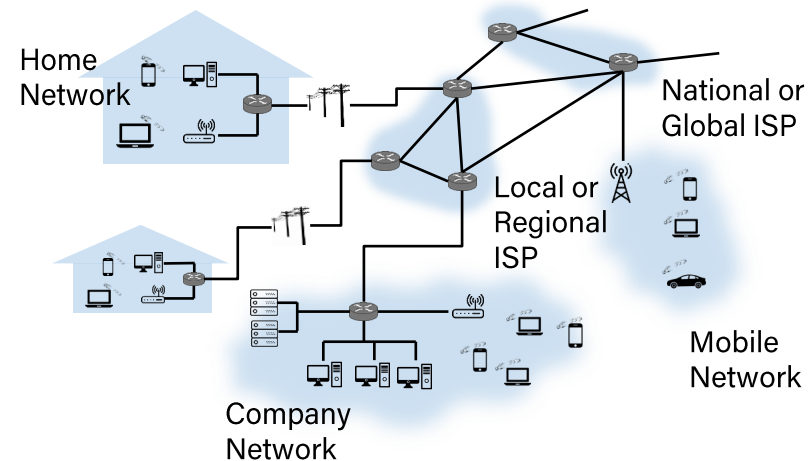


Diagram: Kurose-Ross

Edge of the Internet

- End systems (hosts) e.g., phones, computers, servers...
 - Run or host applications and services

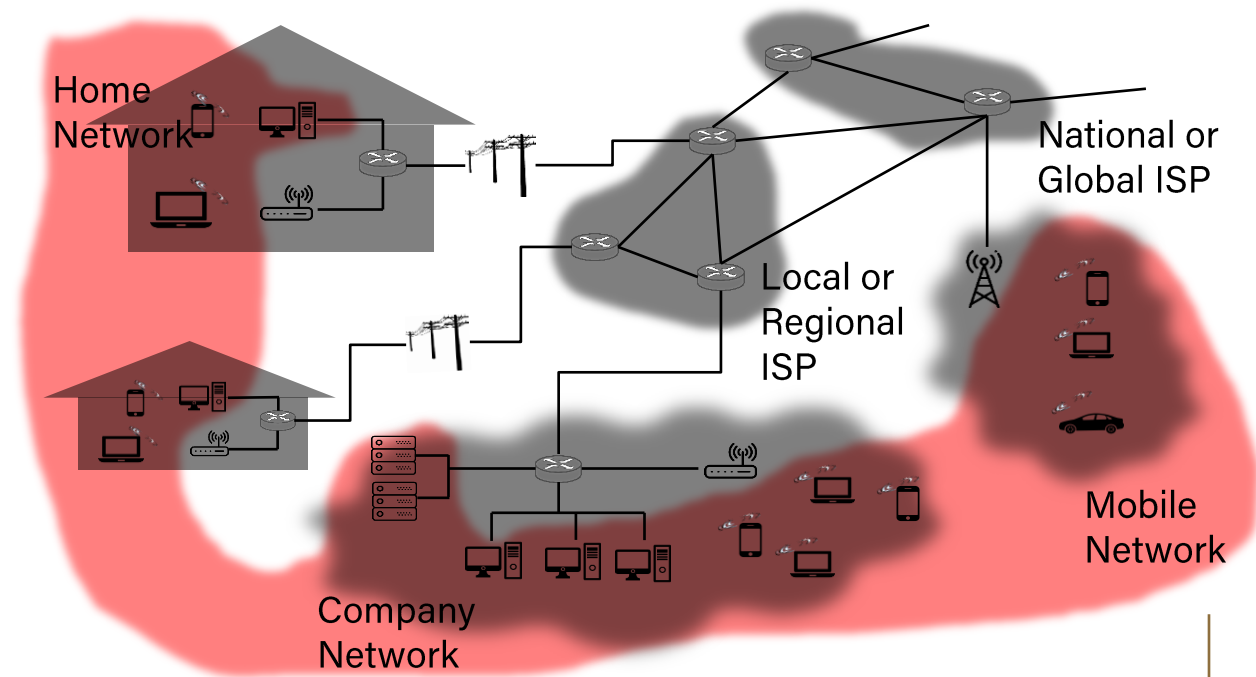


Diagram: Kurose-Ross

Edge of the Internet

- Client-server model: Client sends requests to a server that responds
- Peer-to-peer: Symmetric interaction between hosts (e.g., bittorrent)

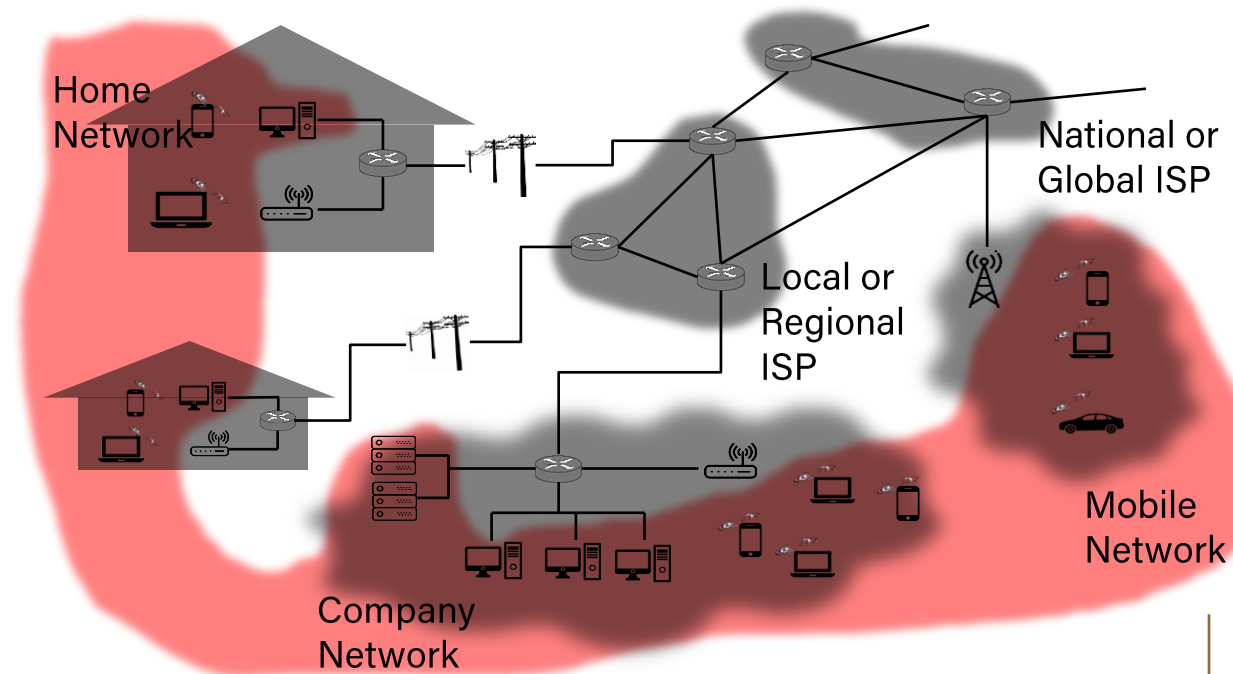


Diagram: Kurose-Ross

Access Networks

- Connections can be wired or wireless
- Symmetric or asymmetric
- Communication links: Fiber, copper, radio, satellite
- Maintained by provider

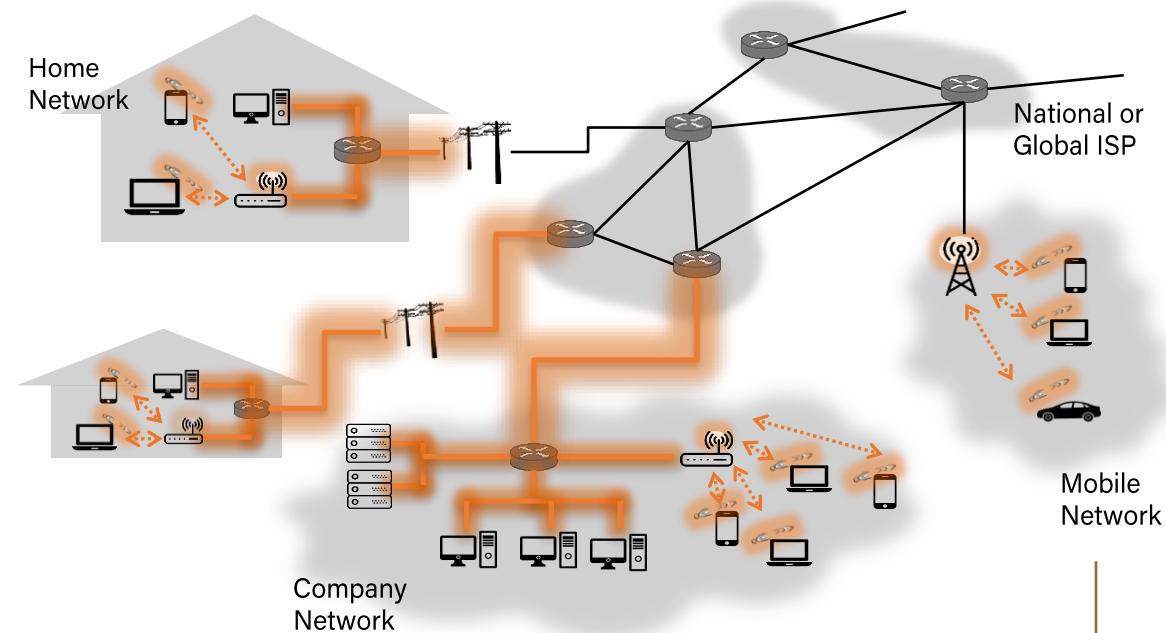
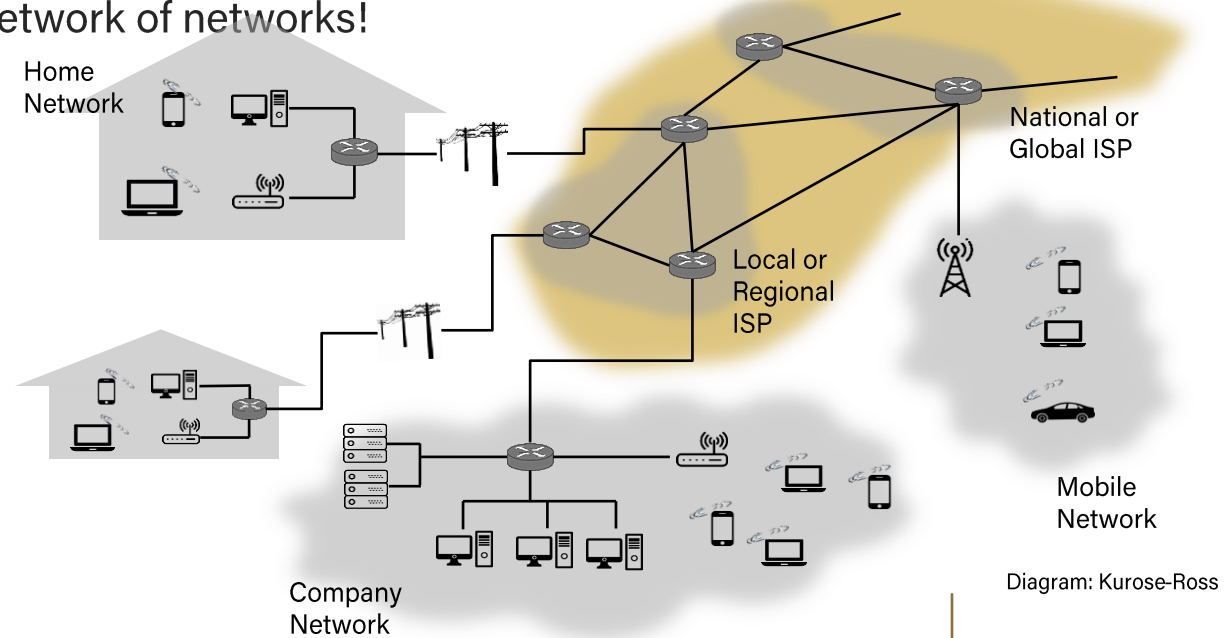


Diagram: Kurose-Ross

Network Core

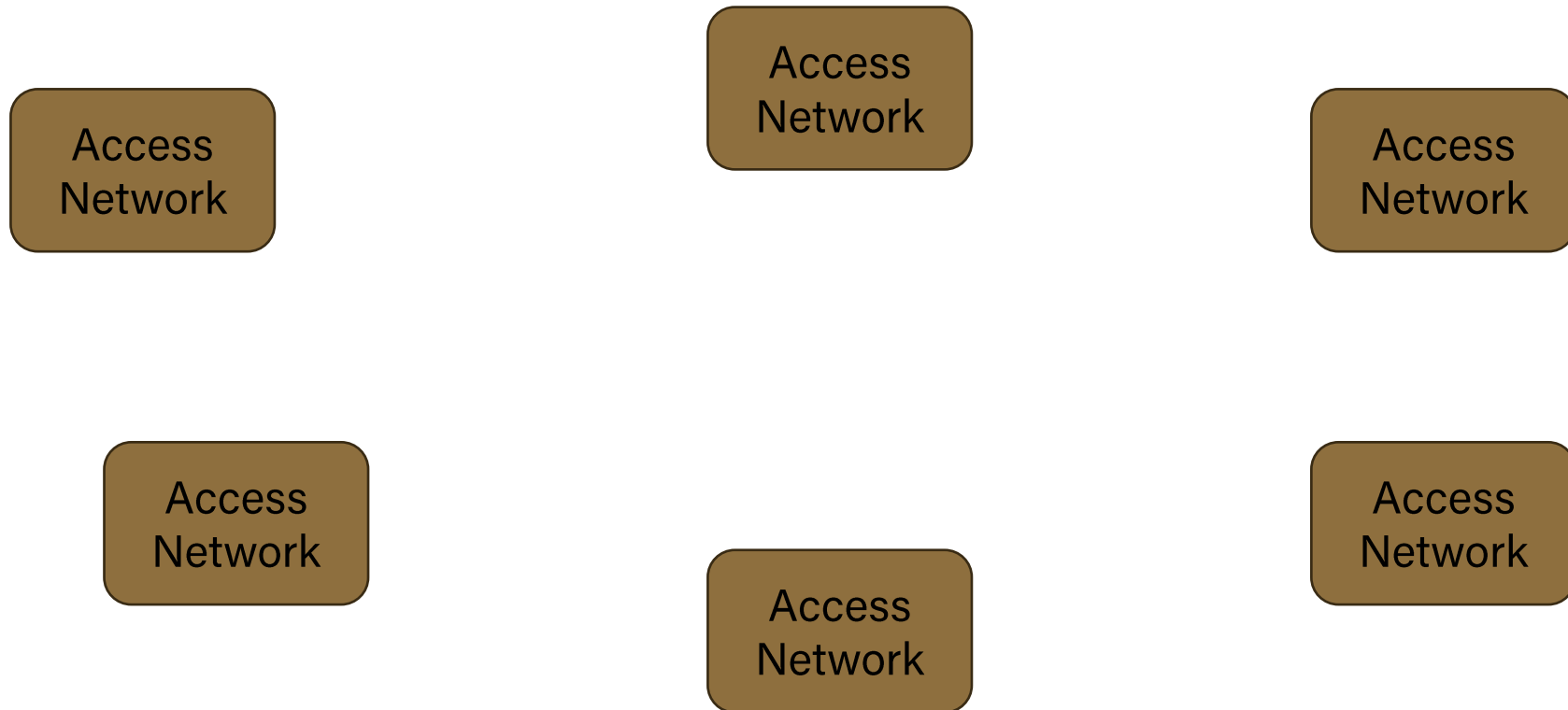
- ISP: "Internet Service Provider"
- e.g. T-Mobile, AT&T, ...
- Hierarchy between ISPs (tier hierarchy)

➔ Internet is network of networks!



Internet: Network of Networks

Why not connect every access network to every other?



Internet: Network of Networks

Why not connect every access network to every other?

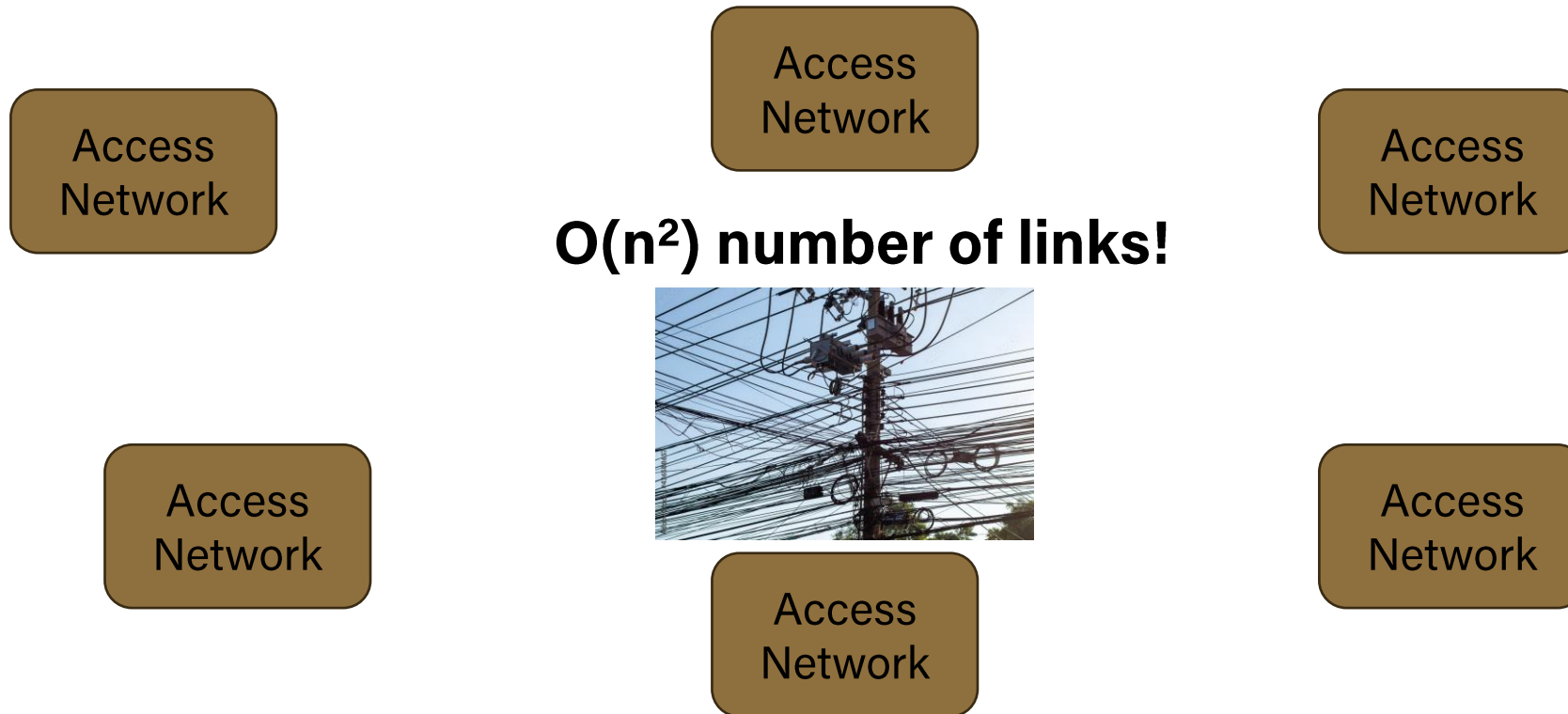
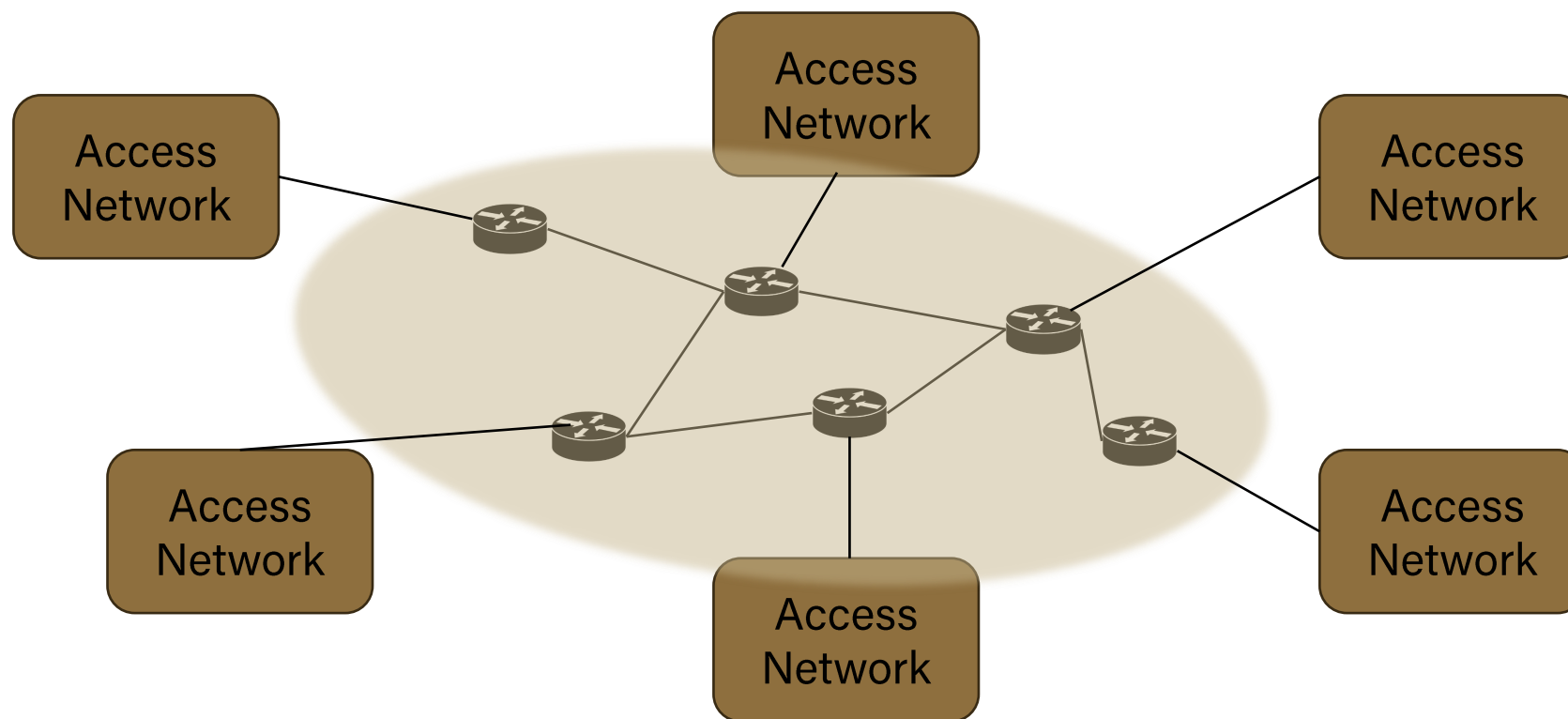


Image: Adobe

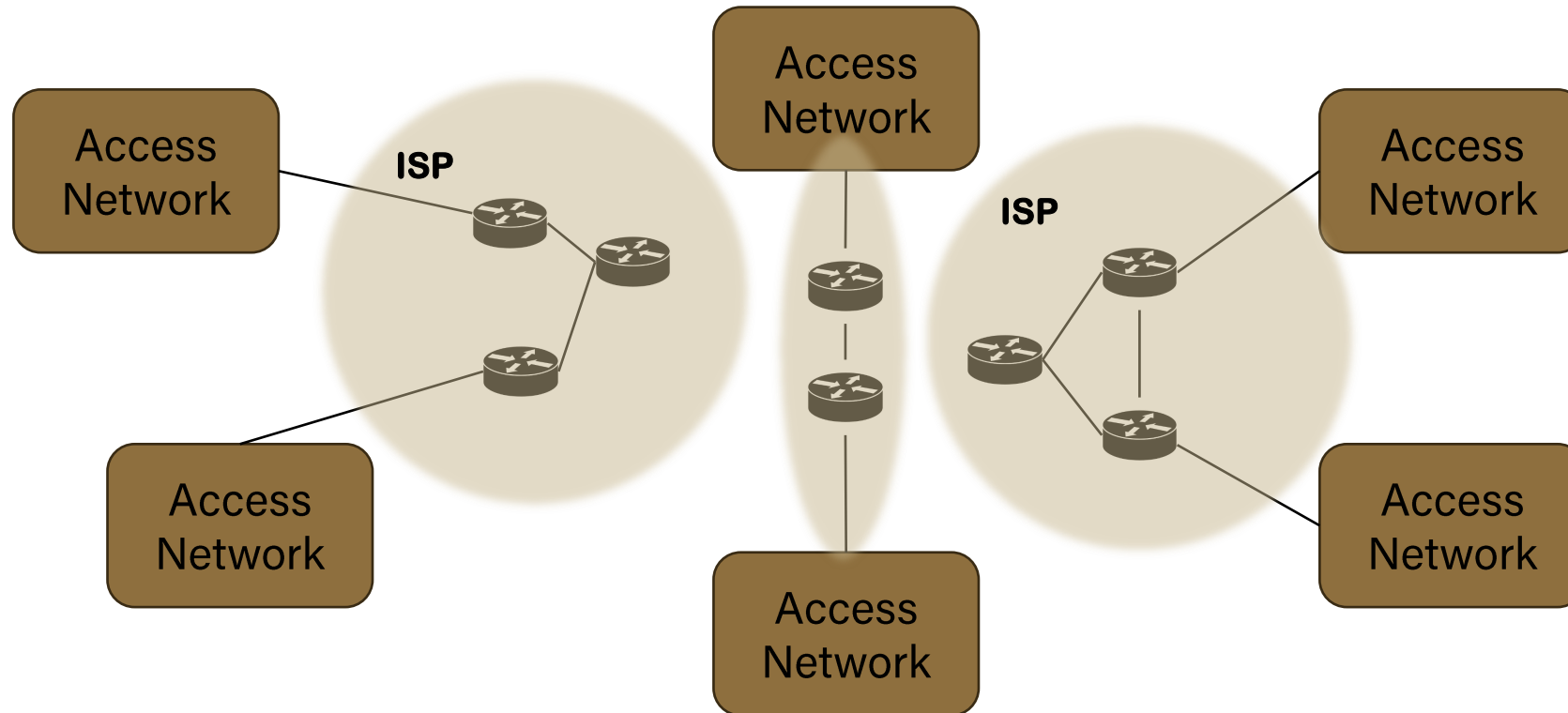
Internet: Network of Networks

A centrally managed network “core”?



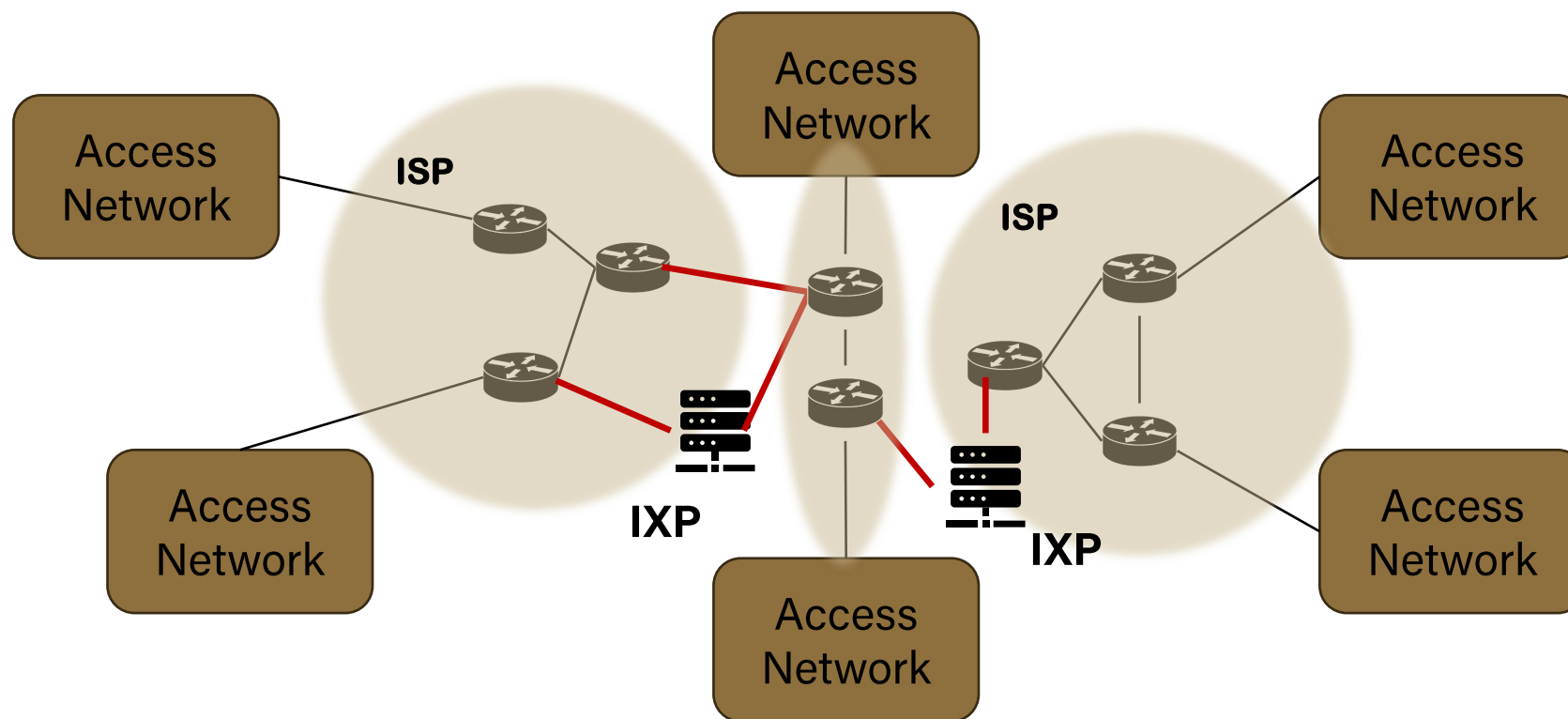
Internet: Network of Networks

Today's internet "core" is a network of networks (ISPs)



Internet: Network of Networks

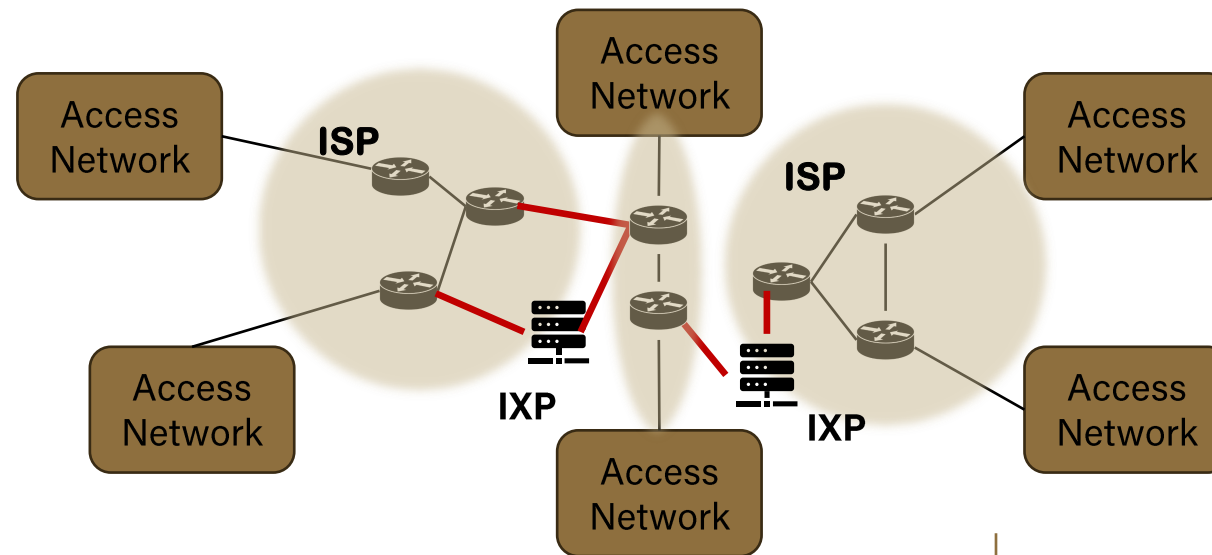
Internet Exchange Points (IXPs)



Internet: Network of Networks

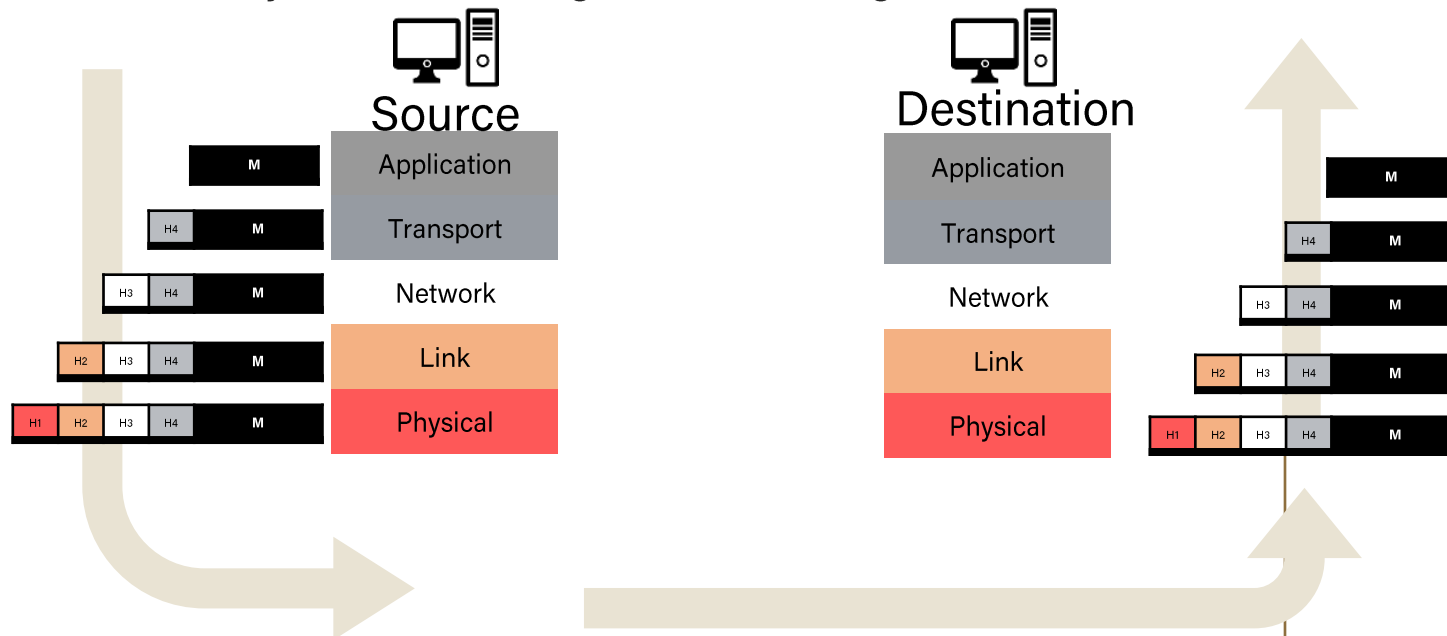
ISPs “peer” with each other using BGP protocol

- Business agreements: I pay you \$, you route my traffic
- Politics
- ...



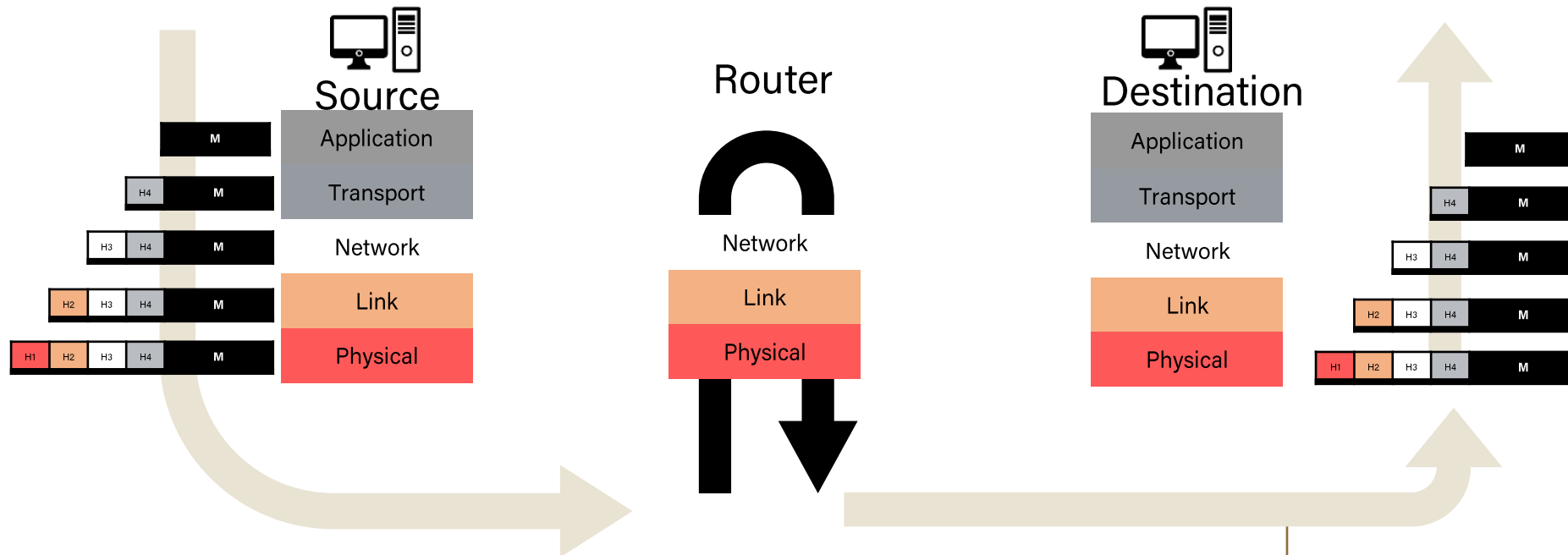
Protocol Layering and Data Transmission

- Message M is sent from top layer
 - Each lower layer adds its header H_n
- Headers contain additional information
 - Necessary for speaking a protocol correctly
 - Generally used for routing and addressing



Protocol Layering and Data Transmission

- Router also removes headers, but only up to the network layer
- Routers, switches and bridges do not implement all layers
 - Intelligence located at the edge of the network

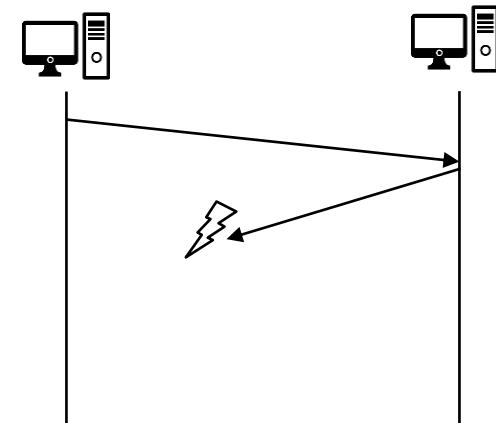


Physical & Link Layer: Network Access

- Physical and link layer enable communication between directly connected parties
- Protocols for this layer
 - Networks: DSL, ISDN...
 - Local networks: Ethernet, Wifi...
 - Point-to-point connections: HDMI, MIDI,...
- Function
 - Transfer bits on the wire
 - Detect bit errors
 - Media access control: “Who is allowed to send and when?”

Network Layer: IP Routing

- Communication over multiple hops
 - Sending packets from source to destination
 - IP addresses as global identifier
 - Source and destination IP in header of network protocol
- Methods for addressing endpoints
 - Unicast (direct communication)
 - Broadcast (send message to everyone on network)
 - Anycast (send message to closest endpoint)
- Problems of the network layer
 - Packet loss
 - Order between messages is not preserved



Transport Layer: TCP and UDP

- Enables data transfer between applications on source and destination machine
 - Allows addressing of single applications
 - Many ports on one machine
 - E.g. port 80 for WWW and port 993 for email
 - Source and destination port numbers in header of transport protocol
- UDP
 - Simple protocol
 - Connectionless
 - Unreliable transport
- TCP
 - Reliable protocol
 - Connection oriented

Application Layer

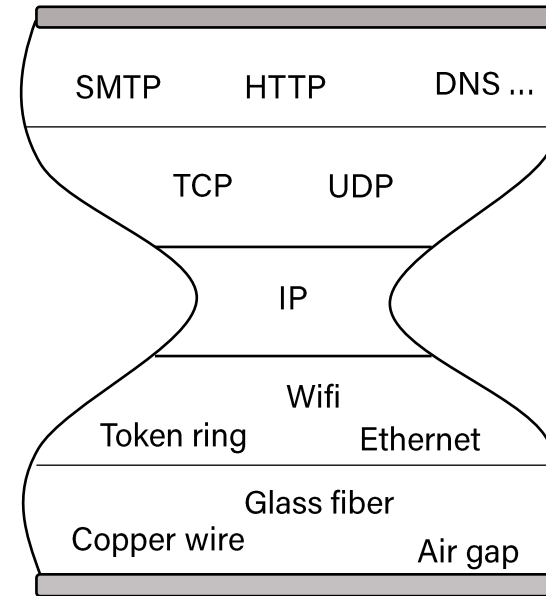
- Applications, e.g., mail Client, browser
 - Receives data over
- Protocols: SMTP, HTTP..
- Relation between WWW and the Internet
 - Uses the HTTP protocol
 - The WWW is a **service** on the Internet
 - Allows people to view web pages
 - But there are many other services!

Who Defines How the Internet Works?

- IETF
 - “Internet Engineering Task Force”
 - Specifies new standards
- IRTF
 - “Internet Research Task Force”
 - Research oriented
- RFCs
 - “Request for Comment”
 - Some of them are Standards for the Internet, others are Best Current Practice, Experimental, or just Informational.
 - Public documents
 - Define how basic protocols work

Why is the Internet so Great?

- Everyone can deploy applications
 - Just needs to be connected
 - Unlike telephone network
- Multi service network
 - Every application feasible over IP
 - IP over any network technology



Internet Design Philosophy

- Network of networks
 - Across multiple technologies
 - Connect existing networks
- Packet switching
 - Sharing of resources
- Minimal assumptions on network capabilities
 - No delivery guarantees
 - But the network will “try” to deliver
- Best effort service
- Overprovisioning
 - To balance out the disadvantages
 - Possible because hardware is cheap and flexible

Internet Design Philosophy

- High reliability
 - When a link is down: Packets take different route
 - Protocol layering
 - Makes it easier for developers to create new applications
 - Encapsulates complexity
 - Intelligence on network edge
 - End systems responsible for reliable communication
 - No state inside the network
- End-to-end paradigm

Circuit Switching & Packet Switching

- Packet switching is one the major reasons for the success of the internet
- Datacenters appear to come back to circuit switching for bandwidth scalability, latency, and various advantages attributed to fully optical switching technologies

Network Core: Packet & Circuit Switching

- Circuit switching: A circuit is reserved through the entire network for a source-destination pair
 - Guaranteed resources
 - Reserved bandwidth and predictable latency
 - Requires connection setup
- Packet switching: Data is transmitted in discrete “chunks” (packets)
 - No guarantee from the network
 - Best effort
 - Buffers accommodate transient traffic bursts

Circuit Switching

Examples

- Telephone networks



Image: [https://en.wikipedia.org/wiki/Telephone_switchboard#/media/File:Photograph_of_Women_Working_at_a_Bell_System_Telephone_Switchboard_\(3660047829\).jpg](https://en.wikipedia.org/wiki/Telephone_switchboard#/media/File:Photograph_of_Women_Working_at_a_Bell_System_Telephone_Switchboard_(3660047829).jpg)

Circuit Switching

Examples

- Datacenter Networks: Optical circuit switching
 - Google's Jupiter
 - TPUv4: Supercomputing cluster with 4096 TPUs

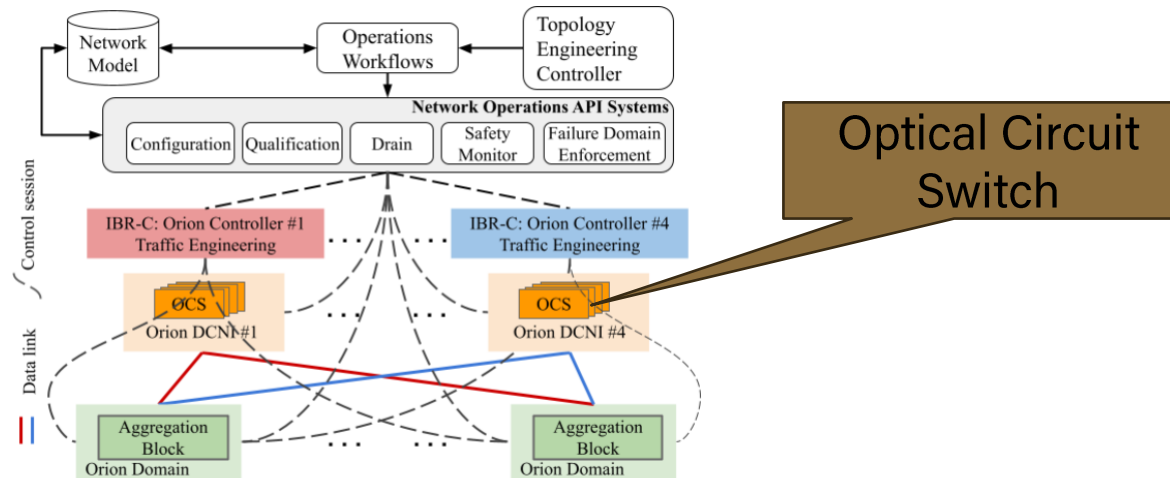
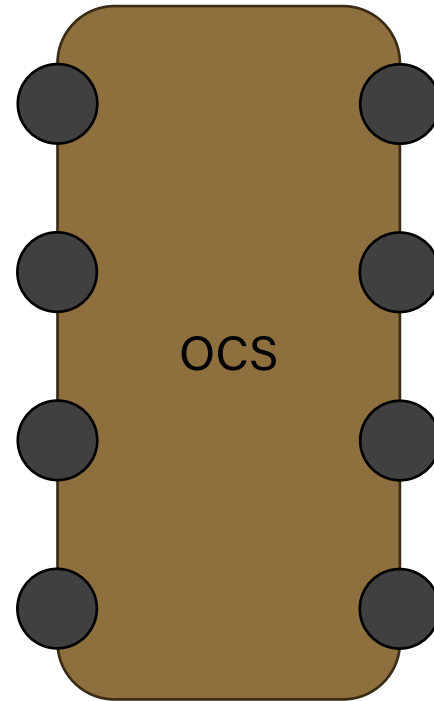
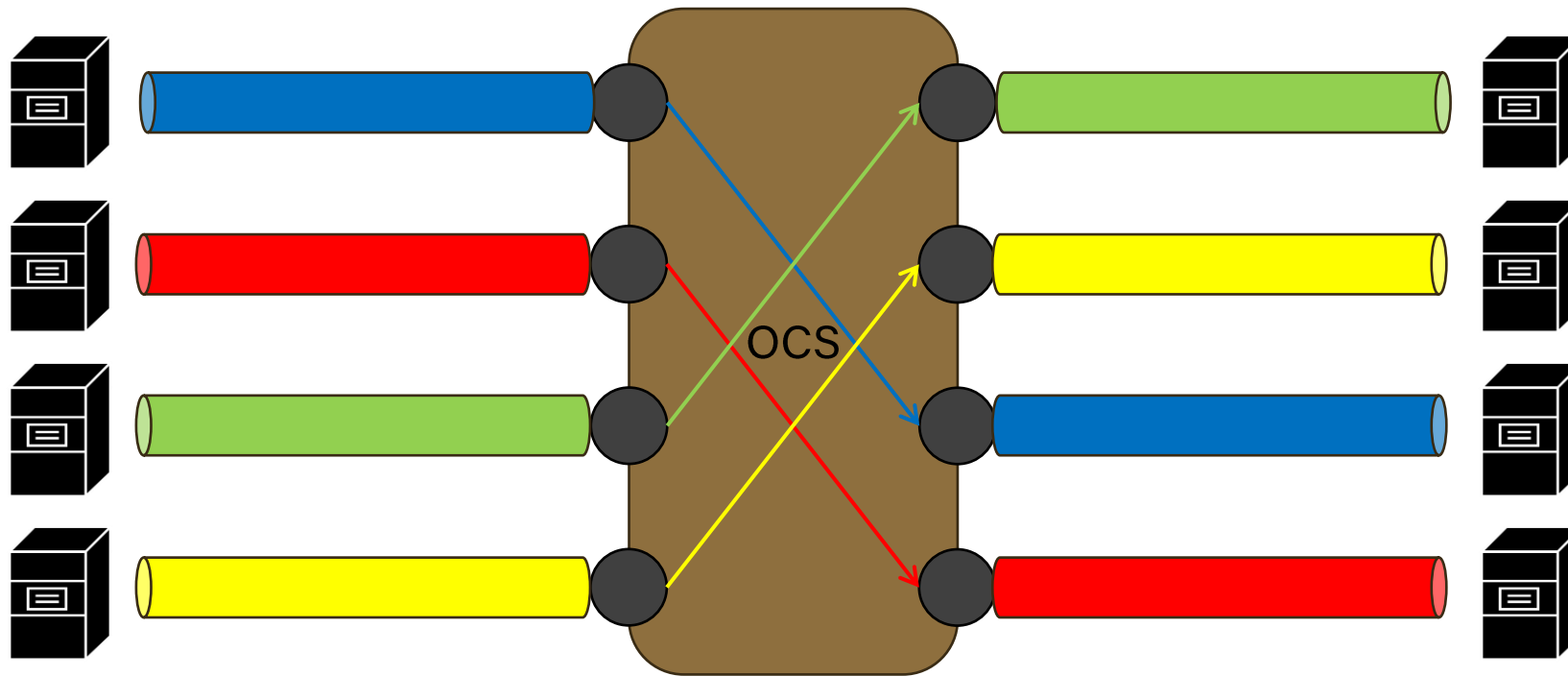


Image: Jupiter Evolving: Transforming Google's Datacenter Network via Optical Circuit Switches and Software-Defined Networking, ACM SIGCOMM 2022.

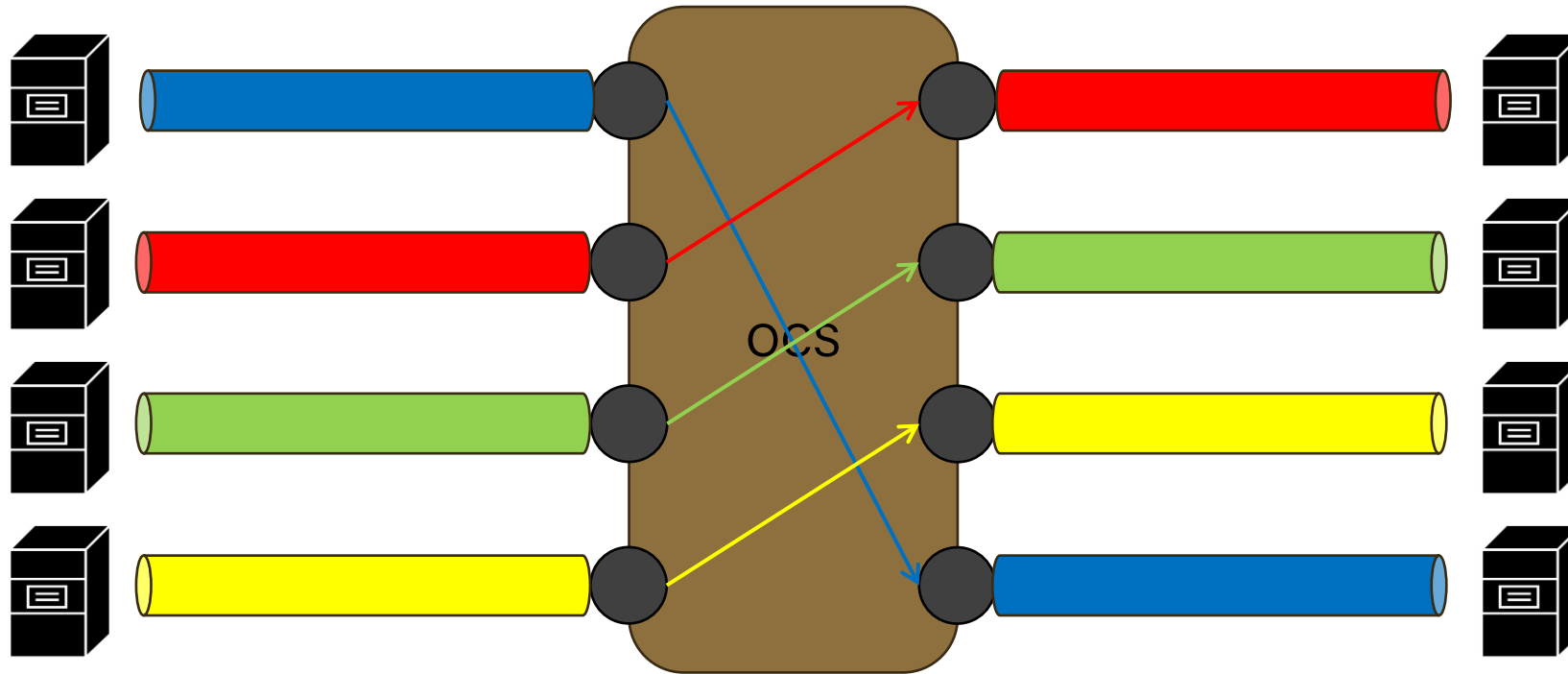
Circuit Switching



Circuit Switching



Circuit Switching

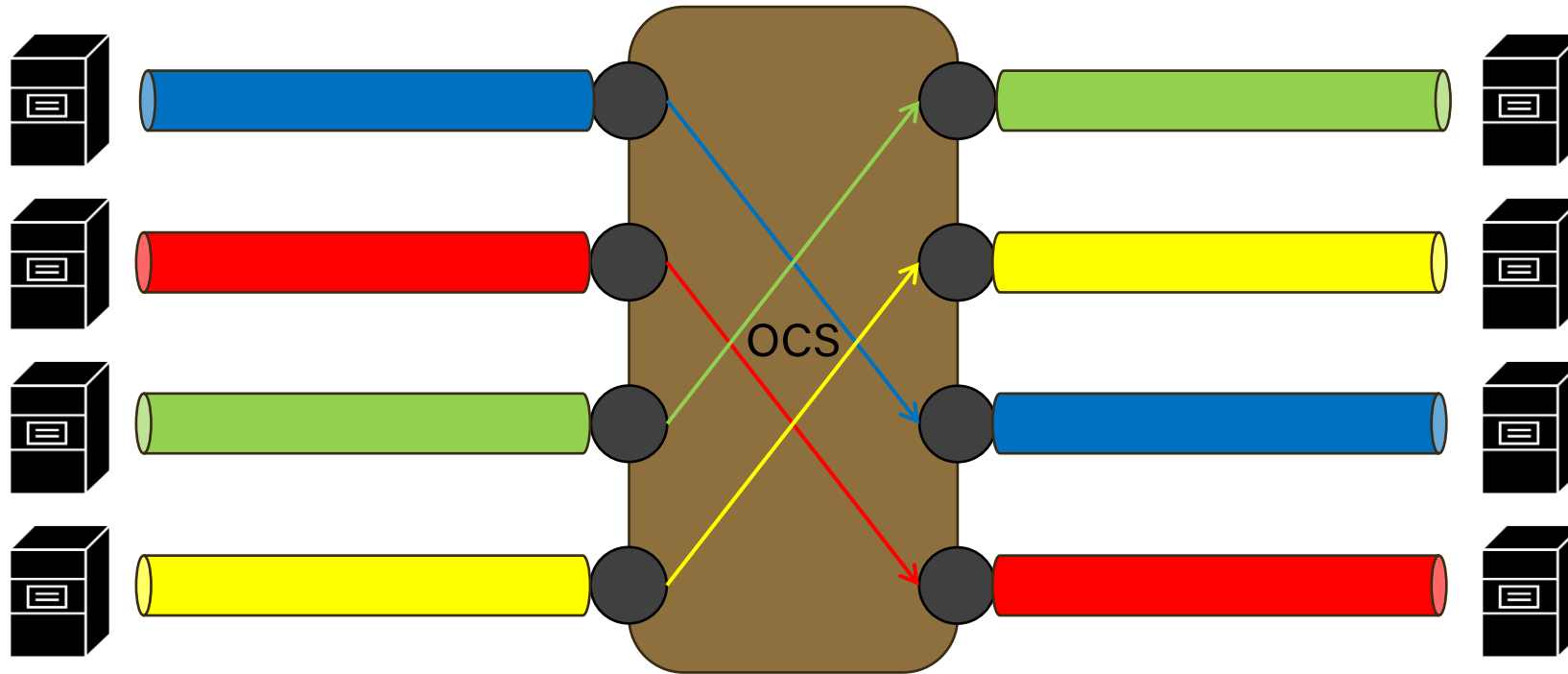


Circuit Switching

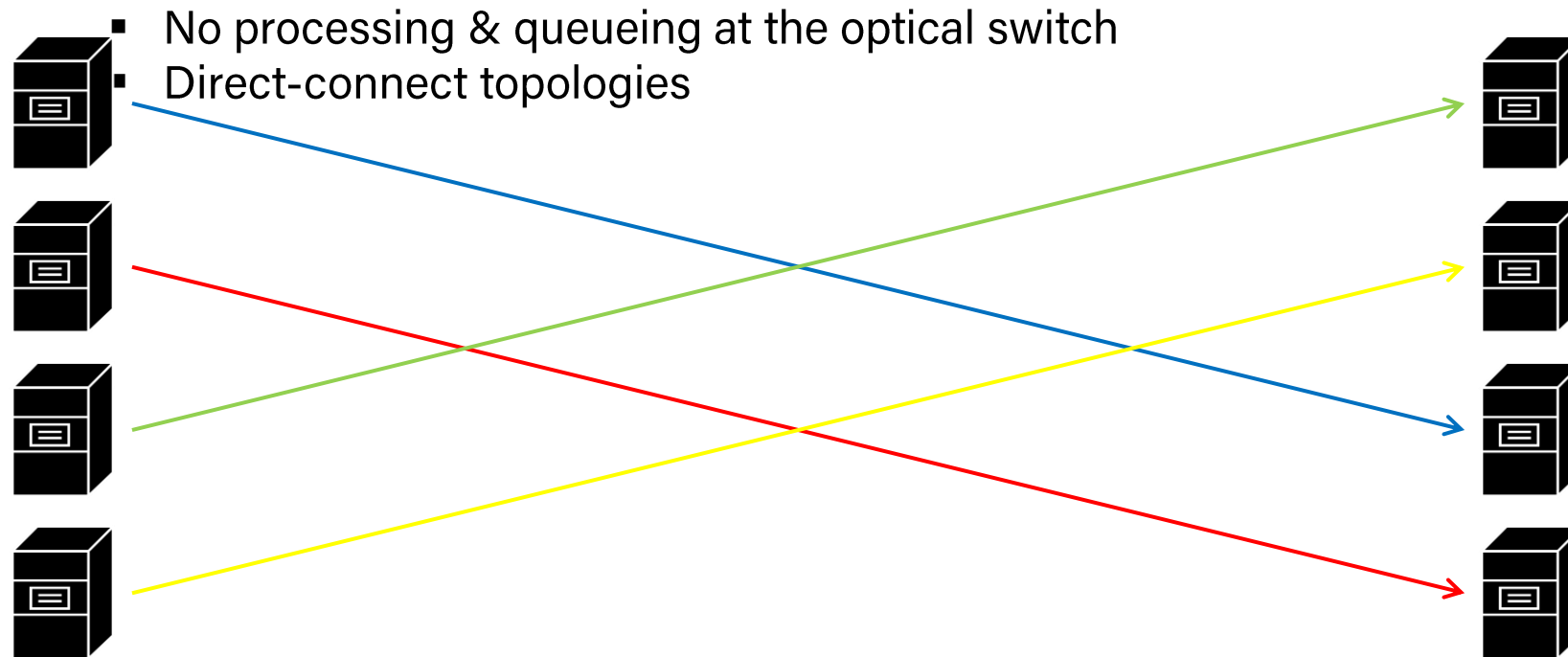
Optical Circuit Switches

- Circuit switch implements a *matching* between input and output ports at any time instance
- Switch can be reconfigured to a different matching
 - Reconfiguration delay
- **No processing:** Simply redirect input signal to output
- **No buffers**
- *More on switching algorithms and topologies later in the course*

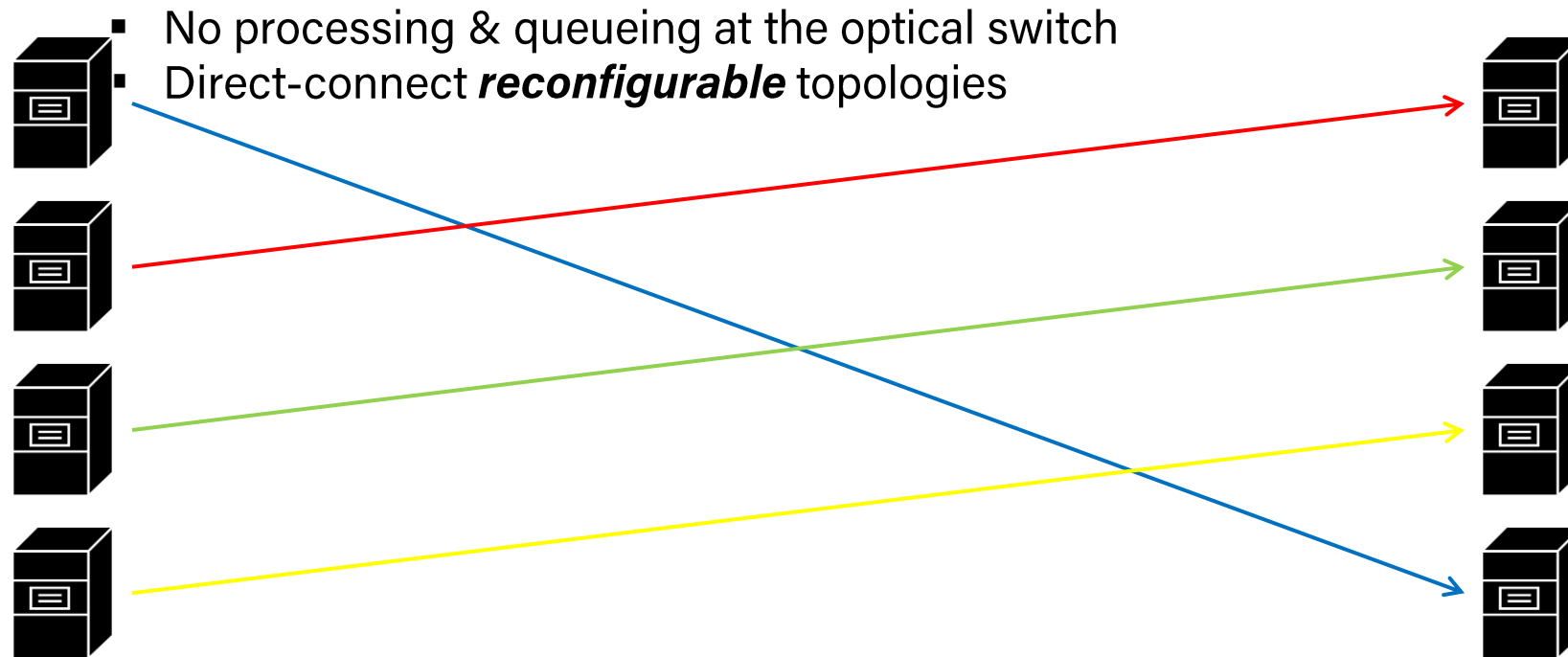
Circuit Switching



Circuit Switching



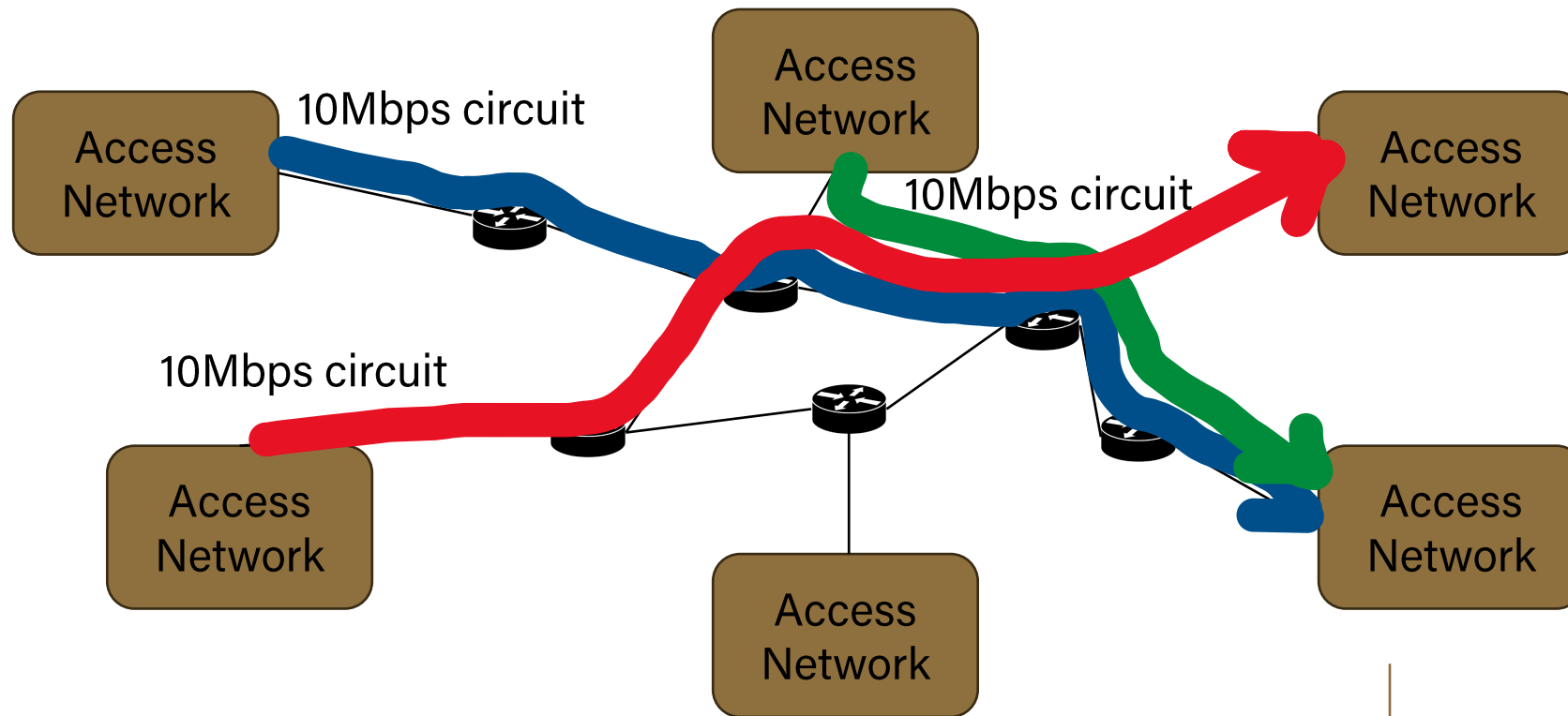
Circuit Switching



Circuit Switching

Virtual Circuits

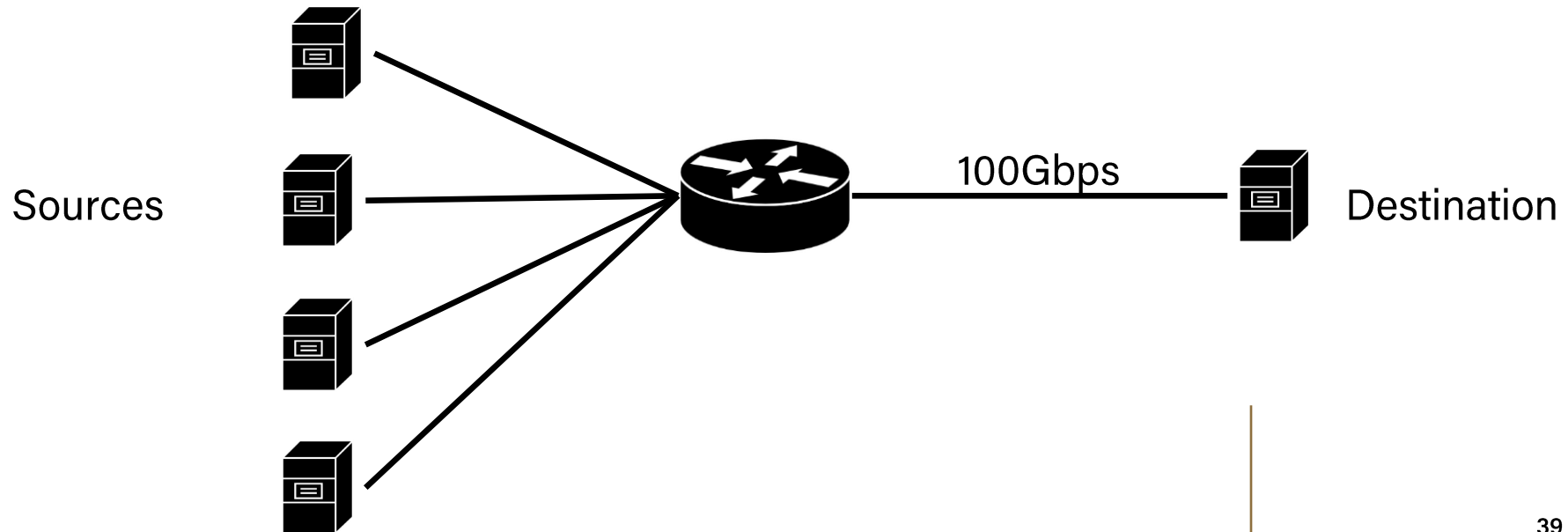
- A circuit-like behavior using resource reservation



Circuit Switching

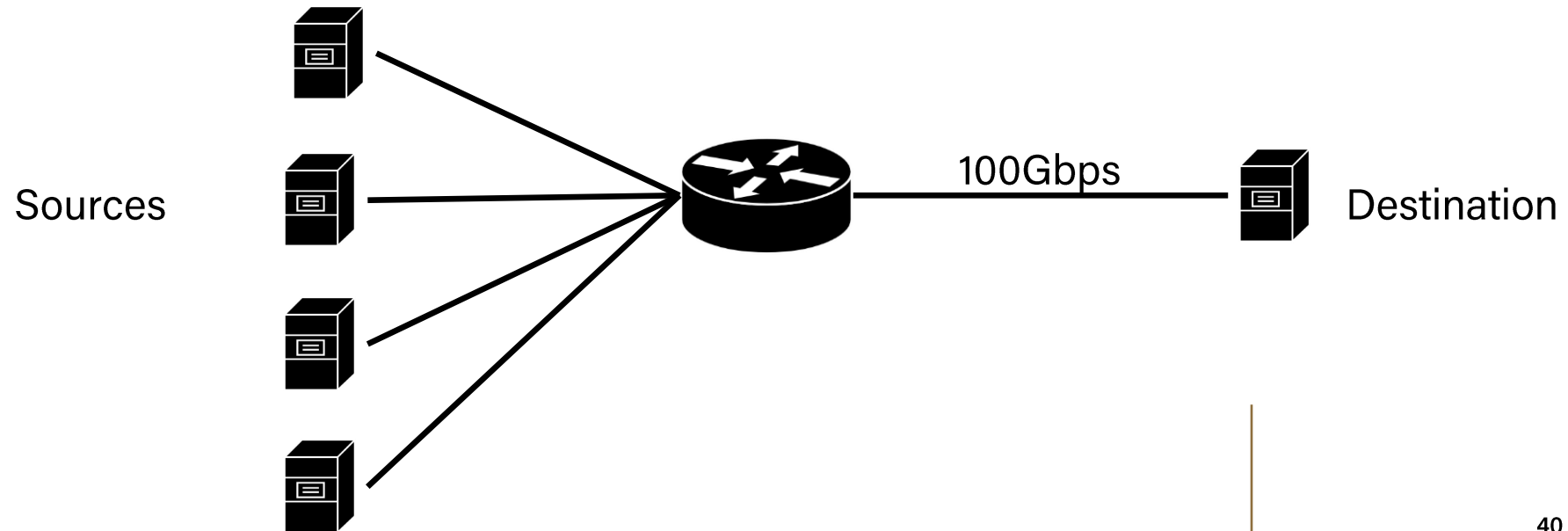
Question

- Each source requires a connection of 100Gbps to the destination but only uses it for $\frac{1}{4}$ fraction of each second.
 - How many connections can this network support, assuming circuit switching
 - What is the average utilization?



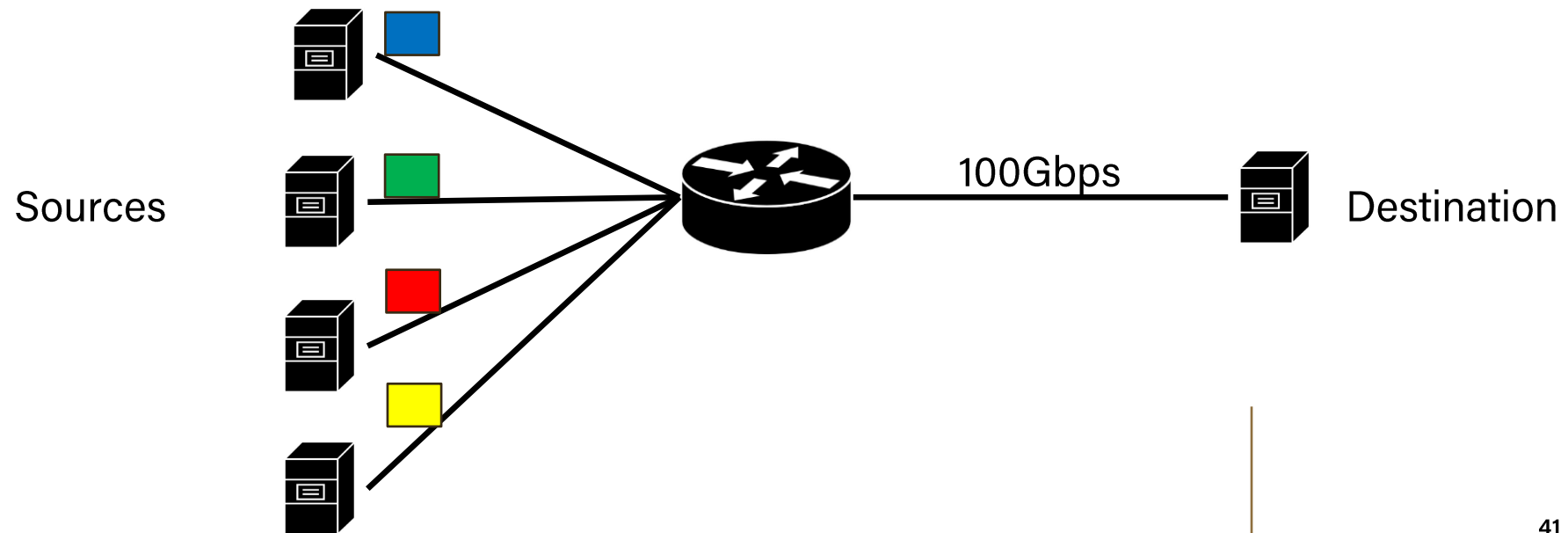
Packet Switching

- Maximum Transmission Unit (MTU) is typically 1500Bytes
 - Jumbo frames 9000Bytes
- Packet processing at each hop: Determine output port, and next hop
- Queueing delay



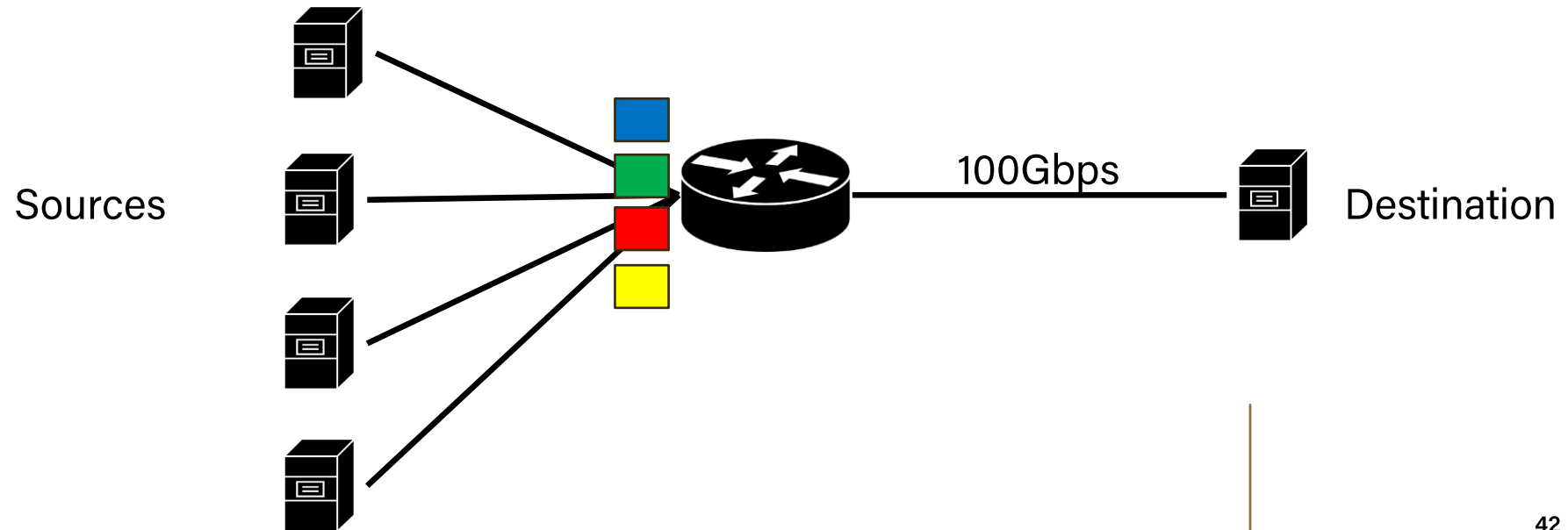
Packet Switching

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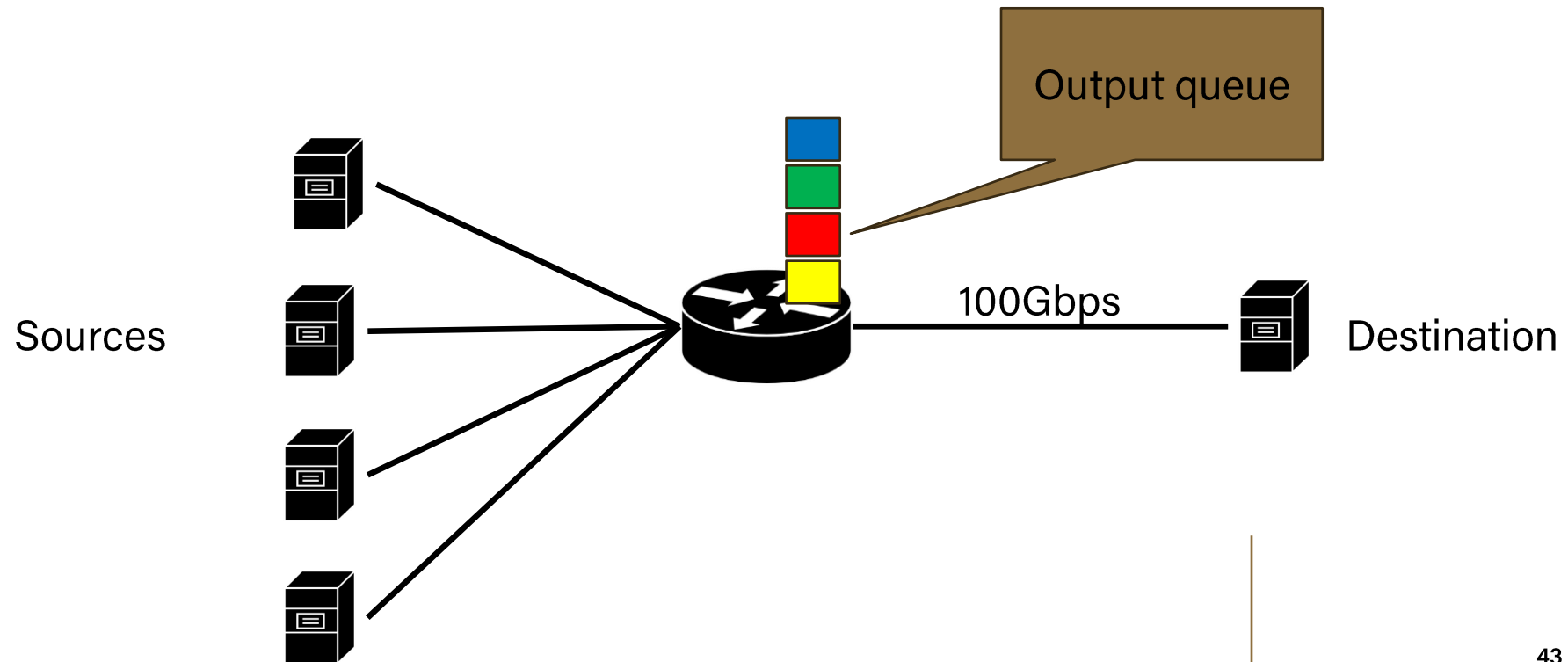
Packet Switching

- Packet processing at each hop: Determine output port, and next hop
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Packet Switching

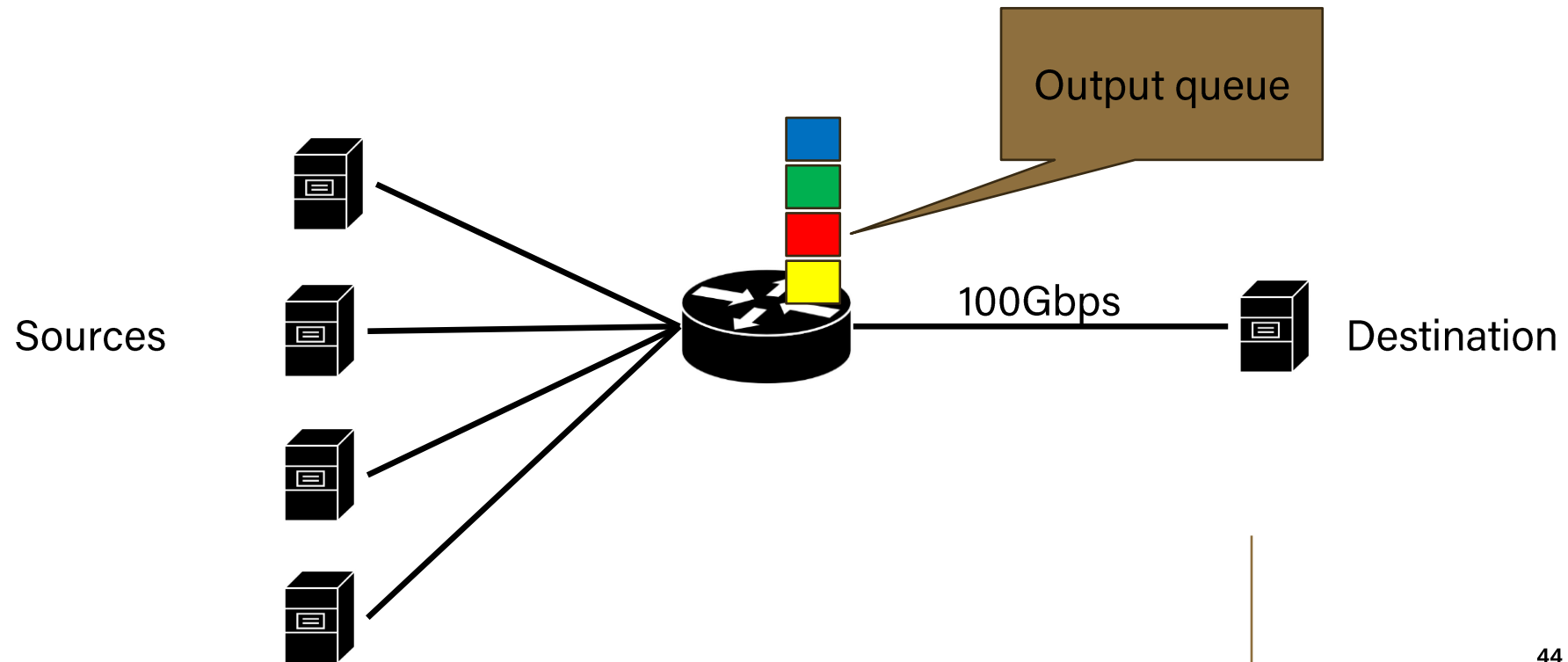
- Packet processing at each hop: Determine output port, and next hop
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Packet Switching

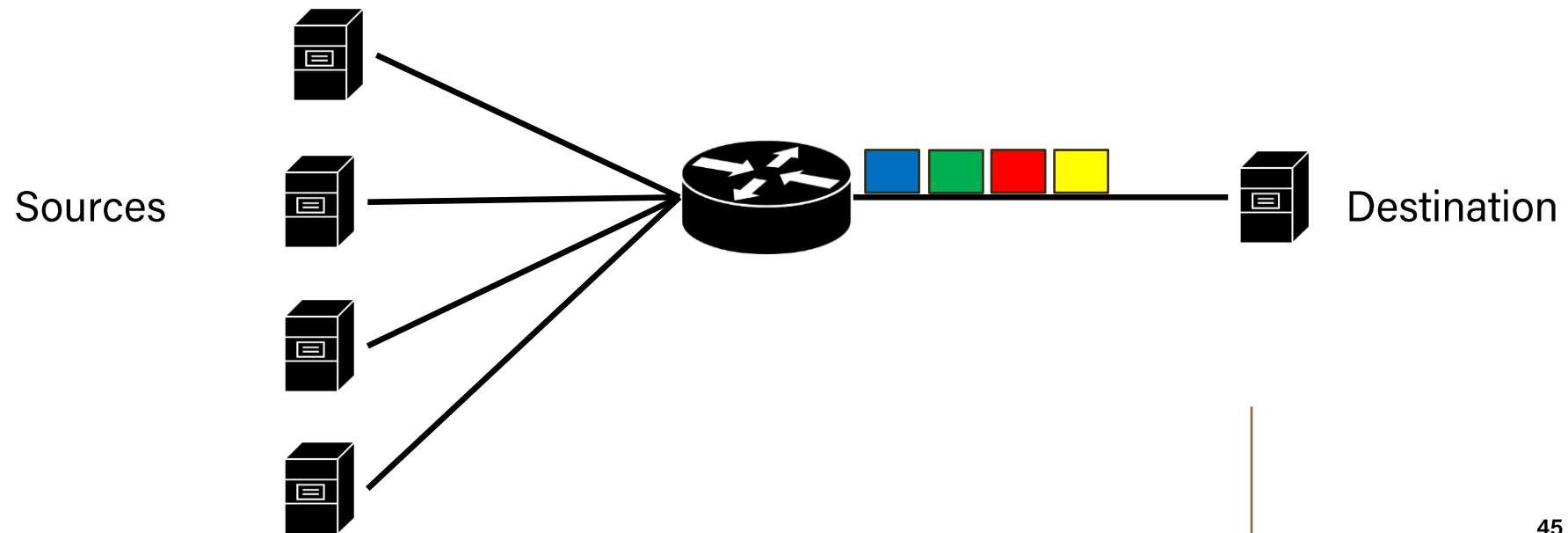
Buffer Allocation and Scheduling

- **Buffer sharing:** Allocate buffer for each queue in the device
- **Active queue management:** Drop or mark each packet (a feedback)
- **Scheduling:** Should blue packet be sent first, or yellow?



Packet Switching

- Packet processing at each hop: Determine output port, and next hop
- Queueing delay



Network Latencies

Transmission Delay

- The time it takes to serialize data
- b : Bandwidth
- s : Total bytes to be transmitted
- Transmission delay is s/b
- Note: s includes the packet headers, where each packet is limited to MTU size
 - Application can inject at most MSS size of “chunks” to the transport layer
 - MSS = maximum segment size, configured by TCP
 - Effectively, MSS size + total header bytes is the total packet size
 - MSS + total header bytes $\leq$ MTU

Network Latencies

Propagation Delay

- Time it takes for 1 bit to travel from one end to the other on a medium
- Typically ~ speed of light in optical fiber ~ 2×10^8 m/sec
- Length of wire dictates the propagation delay
- Speed at which bits travel is *different from and unrelated* to bandwidth
 - 10 km wire and 1 km wire differ by 10x in propagation delays
 - Bits travel at the same speed on both the wires
 - Both wires may operate at the same or different bandwidth
 - Bandwidth is decided by the transceivers at the end points (e.g., RJ45, OSFP)
 - The physical medium has a certain bandwidth that it can accommodate but often greater than the equipped transceivers at the end-points in wired networks

Network Latencies

Processing Delay

- The time it takes to perform packet processing: From the reception of the packet until it is queued for transmission
 - Routing table lookup
 - Firewall
 - Any custom functions implemented by the switch

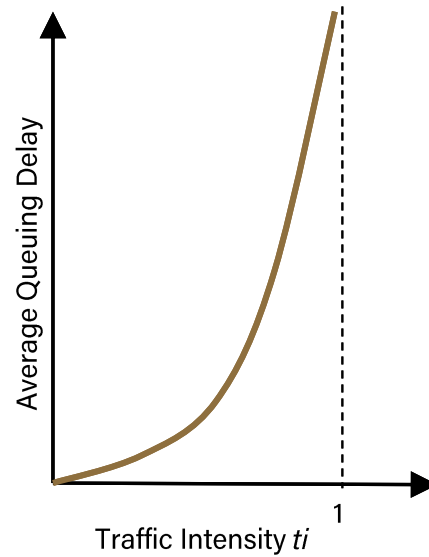
Queueing Delay

- Queueing delay is basically transmission delay in disguise
 - Delay caused by the transmission delay of *other* packets that are in the same queue
 - Time from being queued for transmission after all processing is done, until the packet is fully transmitted out

Network Latencies

Queueing Delay

- Traffic Intensity
 - λ : arrival rate (packets per second)
 - s : packet size (bits)
 - b : bandwidth (bits per second)
 - $T_i = \frac{\lambda \times s}{b}$



Circuit Setup or Reconfiguration Delay

- The time taken to establish a new circuit in the network
 - For virtual circuits, this involves the time from a sender requesting resources, until the circuits are setup
 - For optical switching, this is the time for reconfiguring the circuit switch's matching

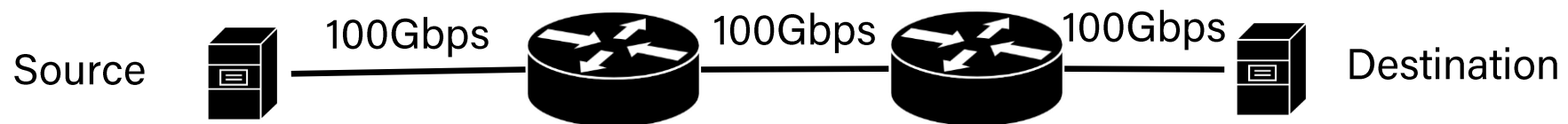
Network Latencies – Store & Forward vs Cut-Through

- Store & Forward: A packet must be fully received to start transmitting the bits
 - Every packet will incur full transmission delay corresponding to packet size *at each hop*
- Cut-through: The router can start transmitting the bits just after reading the destination IP and determining the output port
 - Significantly lower delay
 - Avoids full transmission delay per-packet

Network Latencies

Question

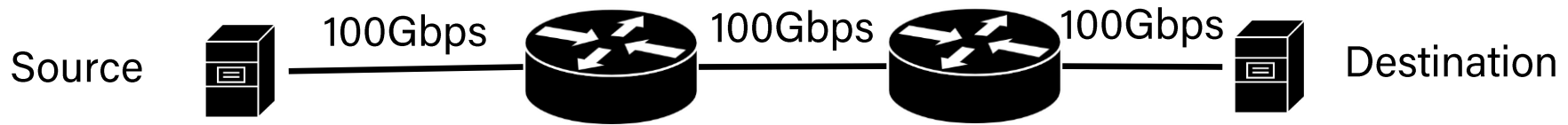
- What is the total time from source to destination to transmit 1250 Bytes amount of data?
 - Every packet incurs a processing delay of 100 nanoseconds
 - MTU = 1250 Bytes (10000 bits)
 - Zero header size
 - Link propagation delay = 1 microsecond



Network Latencies

Question

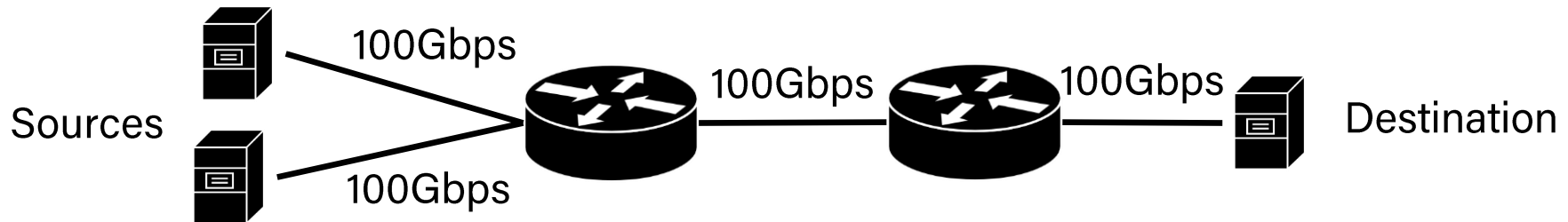
- What is the total time from source to destination to transmit **100 Gb** (10^{11} bits) amount of data?
 - Every packet incurs a processing delay of 100 nanoseconds
 - MTU = 1250 Bytes (10000 bits)
 - Zero header size
 - Link propagation delay = 1 microsecond



Network Latencies

Question

- What is the total time for both the sources to transmit 100 Gb (10^{11} bits) amount of data each to the destination?
 - Every packet incurs a processing delay of 100 nanoseconds
 - MTU = 1250 Bytes (10000 bits)
 - Zero header size
 - Link propagation delay = 1 microsecond



Measuring Network latencies

Ping & Traceroute

- Ping
 - Send an ICMP echo request to a destination IP
 - Destination responds with an ICMP echo reply
 - Difference in time indicates the round-trip delay
 - min value across large number of pings
 - ~ transmission + propagation + processing
 - max value
 - Potentially includes queuing delay
 - or longer paths chosen for certain packets (e.g., due to load balancing)

Measuring Network latencies

Ping & Traceroute

- Traceroute
 - Send an ICMP echo request to a destination IP with increasing TTL
 - TTL = Time-to-Live, an IP header field → Limits the number of hops
 - Intermediate hops respond with ICMP time to live exceeded
 - Difference in time indicates the round-trip delay from source to intermediate hop

THAT'S ALL FOLKS!

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